

Slides: 12. Transactions -
Two Phase Locking.

LECTURE - 26

Wednesday
6/5/26.

Two-Phase Locking: A method to make schedules conflict serializable instead of graphs as it is not practical

- 1) Lock everything at beginning
- 2) Unlock everything at end

⇒ Shared (S) mode: only read lock in this phase & everyone could read

⇒ Exclusive (X) mode: only write lock in this phase but only the one who put the write lock could read & write both; rest everyone has to wait (using queue).

• Since flushing followed by queue; $\therefore O(1)$ time instead of graph (precedence).

Deadlocks:

• Solution 1: Prevent on the first place by keeping a track key graph say cycle to nai ban rha if one edge is added.

• Wound-Wait & Wait-Die: T_2 has already a lock & T_1 wants the lock. Check priority now!

wound wait $\rightarrow$ If T_1 has higher priority; abort T_2 . Otherwise T_1 has to wait for resource

wait-die $\rightarrow$ If T_1 has higher priority; it is allowed to wait otherwise it is aborted. (Nothing for T_2).

$\Rightarrow$ Starvation: when a particular transaction constantly keeps waiting & never gets a turn to be executed.